2501-PYL1003 (Mechanics & STR) — Practice Mid-Sem Paper 3

Full Marks: 50 | Time: 2 hours | Instructor Style: Pradipta Ghosh (IIT Delhi)

Instructions

An MCQ can also be an MSQ (multiple correct options). No credit without justification/derivation.
Show all intermediate steps. Numerical answers without units lose marks.
Approximations must be justified explicitly.

Q1. In a certain system of units, a physicist decides to treat force F, velocity v, and time T as the three fundamental (base) quantities, instead of mass, length, and time.

(i) Express the dimension of mass, length, and (ordinary) time in terms of the new base dimensions $[F], [v], [T]$. **(3)**

(ii) Hence find the dimension of energy in this new system, in terms of $[F], [v], [T]$. **(2)**

Q2 (MSQ). Consider the equation $P + \dfrac{an^2}{V^2} = \dfrac{nRT}{V - nb}$ (the van der Waals equation), where P = pressure, V = volume, n = number of moles, T = temperature, R = universal gas constant. Which of the following is/are correct? **(1+1+1+1=4)**

a. a has dimensions of $[\text{Pressure}] \times [\text{Volume}]^2 \times [\text{mol}]^{-2}$. b. b has dimensions of $[\text{Volume}] \times [\text{mol}]^{-1}$. c. The ratio $a/(Rb)$ has the same dimension as temperature. d. The quantity $\dfrac{a}{V^2}$ must have the same dimension as pressure P for the equation to be dimensionally consistent.

Q3. A particle moves along the x-axis with position given implicitly by the relation $x^2 = 4t^3 + 9$ (SI units, $t \geq 0$).

(i) Find $v(t)$ and $a(t)$ explicitly (as functions of t). **(3)**

(ii) Determine whether the acceleration is ever zero for $t \geq 0$, and if the motion is ever at rest for $t \geq 0$. Justify using your expressions from (i), not by re-deriving from scratch. **(2)**

Q4. A particle of mass m is projected vertically upward with initial speed u in a medium where the air resistance is proportional to the **square** of the speed, with the same proportionality constant k on both the way up and the way down (i.e., resistance always opposes motion).

(i) Write the two separate equations of motion: one valid while the particle moves upward, and one while it moves downward. **(2)**

(ii) Find the maximum height H reached, in terms of m, g, k, u. **(3)**

(iii) Without fully solving the downward motion, argue (using an energy-type argument on the two equations of motion) whether the speed with which the particle returns to the starting point is greater than, less than, or equal to u. **(2)**

Q5. A double block system: block A (mass m_A) rests on top of block B (mass m_B), which rests on a frictionless floor. The coefficient of friction between A and B is μ. A horizontal force F is applied to the **top** block A.

(i) Find the maximum value of F, call it $F_{\max}$, for which A and B move together without slipping, in terms of m_A, m_B, μ, g. **(3)**

(ii) For $F > F_{\max}$, find the acceleration of each block separately. **(3)**

Q6. A point P has spherical polar coordinates $r = 25, \theta = 106.26^\circ, \phi = -36.87^\circ$ (angles rounded to two decimals).

(i) Find the Cartesian coordinates of P (round to the nearest integer). **(3)**

(ii) Find the cylindrical polar coordinates (ρ, ϕ, z) of P. **(2)**

Q7. A curve in three dimensions is described in spherical coordinates by $r = r_0$ = constant and $\phi = \phi_0$ = constant, with θ increasing linearly in time as $\theta = \omega t$ (ω constant). [This is a "great circle" traced at constant angular rate.]

(i) Using $\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}$, find $\vec{v}(t)$ for this motion. **(2)**

(ii) Using $\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2)\hat{\theta} + (\dots)\hat{\phi}$, find $\vec{a}(t)$ for this motion, and show that $|\vec{a}| = r_0\omega^2$ for all t (as expected for uniform circular motion on a great circle). **(4)**

Q8. A vector field is given as $\vec{F} = y\,\hat{\imath} - x\,\hat{\jmath}$ in Cartesian coordinates (this is the classic "circulating" field).

(i) Using $x = \rho\cos\phi, y = \rho\sin\phi$, and the standard relations $\hat{\rho} = \cos\phi\,\hat{\imath} + \sin\phi\,\hat{\jmath}, \hat{\phi} = -\sin\phi\,\hat{\imath} + \cos\phi\,\hat{\jmath}$, express $\vec{F}$ **entirely** in terms of ρ, ϕ and the cylindrical unit vectors $\hat{\rho}, \hat{\phi}$. Show your work rather than quoting the answer. **(5)**

(ii) Comment on what is physically notable about the fact that $\vec{F}$ has **no** $\hat{\rho}$ component after this transformation. **(1)**

Q9. A particle of reduced mass μ moves under a central attractive force $F(r) = -\frac{k}{r^3}$ (an inverse-cube force, **not** inverse-square).

(i) Write the radial equation of motion $\mu\ddot{r} - \frac{L^2}{\mu r^3} = F(r)$, and show that for this particular force law, the equation simplifies to $\mu\ddot{r} = -\frac{k - L^2/\mu}{r^3}$. **(2)**

(ii) Discuss (qualitatively but with reasoning) the three distinct types of radial behavior possible depending on the sign of $k - \frac{L^2}{\mu}$, i.e., whether the particle spirals into the origin, escapes to infinity, or moves at constant r. **(4)**

Q10. A binary star system consists of two stars of masses m_1 and m_2 orbiting their common center of mass in circular orbits, separated by a fixed distance d.

(i) Show that each star moves in a circle of radius $d_1 = \frac{m_2}{m_1 + m_2} d$ and $d_2 = \frac{m_1}{m_1 + m_2} d$ respectively about the center of mass. **(2)**

(ii) Derive an expression for the common orbital period T of both stars, in terms of m_1, m_2, d, and the gravitational constant G (this is the two-body generalization of Kepler's third law). **(4)**

Q11. A block of mass m hangs in equilibrium, supported by two light strings and a pulley system: string 1 goes from the block, over a frictionless fixed pulley attached to the ceiling, and connects to a spring of constant k whose other end is fixed to a wall; string 2 goes from the block straight up to a second point on the ceiling, making an angle α with the vertical, while string 1's segment from the block to the pulley is vertical.

(i) Draw the free-body diagram of the block, and write the two equilibrium equations (horizontal and vertical). **(3)**

(ii) Solve for the tension in string 2 and the extension of the spring, in terms of m, g, k, α. **(4)**

Q12. A particle of mass $0.5\ \mathrm{kg}$ moves in a potential $U(x) = \frac{A}{x^2} - \frac{B}{x}$ for $x > 0$, with $A = 2\ \mathrm{J \cdot m^2}$ and $B = 4\ \mathrm{J \cdot m}$ (this is a 1D model of the effective potential in a central-force problem).

(i) Find the equilibrium position x_0 and determine if it is stable. **(3)**

(ii) Find the minimum energy $E_{\min}$ the particle must have to escape to $x \to \infty$, and describe the motion if the particle has exactly this energy and starts at rest at x_0's... [note: comment on whether it can reach x_0 in finite time if released from rest exactly at the potential minimum]. **(3)**

Q13. Two identical point masses m are connected by a **massless rigid rod** of length $2a$ (not a spring), free to rotate about its center in a horizontal plane (a "dumbbell"), which is itself free to translate. At $t = 0$, a sharp impulsive force J (impulse, in $\mathrm{kg} \cdot \mathrm{m/s}$) is applied perpendicular to the rod at one end only, and acts for a negligibly short time.

(i) Find the velocity of the center of mass of the dumbbell immediately after the impulse. **(2)**

(ii) Find the angular velocity ω of the dumbbell about its center of mass immediately after the impulse (treat each mass as a point mass at distance a from the center). **(3)**

Q14. For a rigid body rotating about a fixed axis with angular velocity $\vec{\omega} = \omega\hat{k}$, a point mass m is located at position $\vec{r} = (x, y, 0)$ relative to a point O on the axis.

(i) **Prove**, starting from $\vec{v} = \vec{\omega} \times \vec{r}$, that the kinetic energy of this mass can be written as $\frac{1}{2}m\omega^2(x^2 + y^2)$. **(2)**

(ii) Hence, for a system of N such masses rigidly rotating about the same fixed axis, **prove** that the total kinetic energy equals $\frac{1}{2}I\omega^2$, where $I = \quad {}_i\, m_i(x_i^2 + y_i^2)$ is the moment of inertia about that axis. State this derivation cleanly as a proof, not just a statement of the formula. **(3)**

Q15. A $3\ \mathrm{kg}$ block is connected to a $2\ \mathrm{kg}$ block via a light string over a frictionless pulley at the top of a frictionless double incline (a symmetric "wedge" with two inclined faces at angles 30° and 53° to the horizontal, meeting at the top where the pulley sits). Block A $(3\ \mathrm{kg})$ is on the 30° face; block B $(2\ \mathrm{kg})$ is on the 53° face. Take $\sin 53^\circ \approx 0.8$, $\cos 53^\circ \approx 0.6$.

(i) Determine the direction of impending motion (which block slides down) by comparing the components of gravity along each incline. **(2)**

(ii) Find the acceleration of the system and the tension in the string, using $g = 10\ \mathrm{m/s}^2$. **(3)**

(End of Paper 3 — 50 marks)